

Mahmoud Elsharawy

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EDUCATION

Cornell University

Masters of Engineering in Computer Science

Bachelors of Science in Computer Science & Electrical and Computer Engineering

GPA: 3.78

May 2025

Dec 2024

EXPERIENCE

Apple

System Electrical Engineer

July 2025 - Present

Cupertino, CA

- Integrated schematics for PCBA development, coordinating with Mechanical Engineers, Design for Manufacturing, and PCB Designers to create production-worthy PCBAs
- Characterized and debugged hardware using oscilloscopes and signal analyzers to ensure reliability in production
- Developed a Python tool to automate the creation of system interconnect diagrams by fetching existing board geometry from ECAD libraries and using ML techniques to cleanly orient and position them in an image file

Apple

System Electrical Engineering Intern

May 2024 - Aug 2024

Cupertino, CA

- Developed a Python tool to automate HSPICE simulations for existing Cadence designs and GPIO specifications

Apple

System Electrical Engineering Intern

Jan 2023 - Aug 2023

Cupertino, CA

- Wrote firmware in C for an SAMD21 microcontroller on a custom PCBA to calibrate its ADC and control a precise micro-current load to characterize power components
- Designed and tested a buck converter power module for use on internal dev boards, removing reliance on a vendor's power modules, preventing future supply chain issues and reducing cost

SpaceX

Hardware Engineering Intern

Jan 2022 - Aug 2022

Hawthorne, CA

- Created a Python package to use SCPI commands and REST API interaction to collect and model hardware data
- Automated an assembly line station through mechanical design and PLC TwinCAT software, tripling its speed
- Tested and qualified alternative integrated circuits for Business User Terminals, preventing a parts shortage

PROJECTS

RP2040 Audio Communication | C++, Embedded Systems, Microcontrollers

Mar 2024 - April 2024

- Developed firmware for a 2-way audio communication protocol on Raspberry Pi RP2040 boards using frequency-based bit encoding with Direct Digital Synthesis, enabling arbitrary ASCII character data transmission via sound
- Implemented automatic threshold detection and PLL synchronization logic for each board to detect when the other board starts transmitting and adjust its clock accordingly

Discord Hard Drive | C++, REST API, Linux, Git

Feb 2025 - Mar 2025

- Developed a Discord bot in C++ that interfaces with Discord's REST API to store arbitrary data through reading, writing, and modifying messages
- Created an nbdkit plugin in C++ to integrate the bot with a virtual block device, enabling seamless read/write operations and allowing Discord messages to be used as a hard drive

Chess AI | Rust, Python, REST API, Systemd, Git

Aug 2021 - Present

- Developed a UCI Chess AI in Rust using a minimax algorithm with move-ordering and variable depth control
- Implemented Zobrist hashing with a hash-table to look up previously seen positions, reducing computation time
- Hosted a systemd service to interact with lichess.com's REST API, allowing the AI to play against other bots and humans, earning an Elo rating of 1850, making it stronger than 85% of human players on the website

TECHNICAL SKILLS

Languages: C/C++, Rust, Java, Python, ARM, RISC-V

Developer Tools: CMake, Git, Linux, Nix/NixOS, Cargo